

How I treat chemotherapy-induced thrombocytopenia with thrombopoietin receptor agonists

Hanny Al-Samkari^{1,2} and Gerald A. Soff³

¹Division of Classical Hematology, Mass General Brigham Cancer Institute, Massachusetts General Hospital, Boston, MA; ²Harvard Medical School, Boston, MA; and ³Division of Hematology, University of Miami/Sylvester Comprehensive Cancer Center, Miami, FL

Chemotherapy-induced thrombocytopenia (CIT) is a common complication of cancer therapy that may result in major bleeding events and reduction of chemotherapy relative dose intensity. Although there remain no widely available licensed therapies to manage CIT, major studies, including global pivotal phase 3 clinical trials of the thrombopoietin receptor agonists (TPO-RAs) romiplostim and avatrombopag in CIT, have been completed, along with additional supporting evidence from other clinical trials and observational studies. As with patients with chemotherapy-induced neutropenia, not all patients with CIT require treatment, but for those who do, it can have a major impact on a patient's oncologic care and outcomes. Proper management of CIT requires an understanding of the 2 clinical subtypes of CIT (persistent and nadir), the limited role of platelet transfusions and antifibrinolytics, and when and how to appropriately use TPO-RA support. In this article, 4 patient cases of CIT are used to illustrate common and challenging scenarios occurring in clinical practice. Built on a framework of published studies and supported by extensive clinical experience, appropriate treatment strategies for CIT are described in detail, with emphasis on proper selection of patient, dosing, titration, and monitoring for the use of TPO-RAs in CIT.

Introduction

Chemotherapy-induced thrombocytopenia (CIT) is a common and challenging clinical problem in patients with cancer.¹⁻⁴ CIT is broadly defined as thrombocytopenia caused or exacerbated by traditional cytotoxic and myelosuppressive chemotherapeutics and targeted antineoplastic agents^{5,6} (but not immunotherapies). When significant, CIT may precipitate bleeding or increase bleeding risk and therefore prompts dose reductions and treatment delays of essential antineoplastic treatment.

Major bleeding is a highly morbid complication in the patient with cancer, resulting in emergency visits and hospitalizations, declines in performance status, and halting or discontinuation of antineoplastic therapy. Accordingly, it is associated with rather significant reductions in overall survival.⁷ Unlike most other patients with hypoproliferative thrombocytopenia, such as those with aplastic anemia for whom the risk of spontaneous bleeding is low at platelet counts of $>10 \times 10^9/L$ to $20 \times 10^9/L$,⁸ the risk of spontaneous bleeding in patients with solid tumors is elevated at platelet counts of $>20 \times 10^9/L$.⁹ Patients with cancer and thrombocytopenia are at increased risk of clinically significant and major bleeding at higher platelet count ranges, compared with patients with thrombocytopenia without cancer, whether on anticoagulation or not. Published studies have found an

increased risk of such bleeding at platelet counts of $30 \times 10^9/L$ to $40 \times 10^9/L$ in patients with cancer not receiving anticoagulation and at platelet counts $<100 \times 10^9/L$ in those receiving anticoagulation.^{7,10}

The increased bleeding risk in patients with cancer reflects many concomitant bleeding risks not present in other thrombocytopenic conditions. Aggressively growing solid tumors can invade and disrupt blood vessels, mucosal surfaces, and other tissues with friable, highly vascularized neoplastic tissue. These patients have among the highest risk of thromboembolism of any patient population and therefore commonly receive antithrombotic therapy for primary prevention, secondary prevention, or treatment of thrombosis.^{11,12} They receive radiation therapy, which reduces tissue integrity and can cause bleeding. They often have indwelling catheters and other hardware and commonly require both minor and major surgery. Finally, their nutritional status is often suboptimal, and they are commonly subject to polypharmacy, including consumption of drugs and dietary supplements that impair platelet function.^{13,14}

To avoid highly morbid major bleeding, and in the absence of any widely available licensed therapy to manage CIT, hematologists and oncologists are forced to reduce the dose of chemotherapy, delay chemotherapy treatment, or even discontinue agents or regimens altogether. This negatively affects

the relative dose intensity (RDI) of chemotherapy. The administration of indicated chemotherapy at the studied, intended doses is a foundational principle of medical oncology. Although the impact of RDI reduction exclusively due to CIT on oncologic outcomes has not been well studied, even modest RDI reductions due to any cause have been demonstrated to reduce overall survival, progression-free survival, and likelihood of cancer cure across numerous studies (primarily prospective in the early stage setting and retrospective in the advanced/metastatic setting).¹⁵⁻²² Accordingly, preservation of full RDI with supportive therapies, such as granulocyte colony-stimulating factors for chemotherapy-induced neutropenia, has been demonstrated to improve survival across many studies.^{23,24} Therefore, when possible, avoidance of RDI reduction from CIT is warranted.

Given their broad application to many other etiologies of thrombocytopenia and the lack of approved agents for CIT, the safety and efficacy of multiple thrombopoietin receptor agonists (TPO-RAs) have been extensively evaluated in this setting. Although still off-label, the wide body of evidence for the use of romiplostim,²⁵⁻³⁰ a subcutaneously injected peptibody TPO-RA, in the management of CIT has resulted in its integration into CIT treatment guidelines from the National Comprehensive Cancer Network and International Society on Thrombosis and Haemostasis.^{2,31} Although less data are available regarding the use of oral TPO-RAs to treat CIT, the oral small-molecule drug avatrombopag has a good safety record and emerging efficacy data^{32,33} that have made it an additional option to manage CIT.

Using a series of cases, this article provides guidance on the management of CIT across many clinical scenarios. Most critically, it illustrates scenarios in which treatment is and is not warranted, and when it is warranted, how it is best used. Given the growing role of TPO-RAs for CIT, special emphasis is given to proper patient selection, dosing, titration, and monitoring for these agents in CIT.

Classifying CIT and adjudicating bleeding risk

CASE 1: NADIR CIT

A 62-year-old woman with advanced ovarian cancer receiving combination chemotherapy with carboplatin, paclitaxel, and bevacizumab presents for an urgent visit midcycle during cycle 3 (day 14 of 21) to address management of excessive chemotherapy-related nausea and vomiting. Laboratory evaluation reveals an incidentally discovered platelet count of $50 \times 10^9/L$, which raises concerns given her ongoing therapeutic anticoagulation for atrial fibrillation. During evaluation, she notes a mild nosebleed the previous day in the setting of vomiting that lasted for 5 minutes, as well as a bruise on her thigh from bumping into her bedpost 4 days ago. Review of her medication list also reveals that she is using turmeric and fish oil supplements for arthritis symptoms. One week later, on day 1 of cycle 4, her platelet count had recovered to $168 \times 10^9/L$.

There is no established definition of CIT. We define CIT as thrombocytopenia caused by cytotoxic chemotherapy or

antineoplastic targeted therapy sufficient to increase bleeding risk or impair safe administration of indicated anticancer therapy at full dose and on schedule. Thrombocytopenia resulting from cancer immunotherapies, such as immune checkpoint inhibitors, is not CIT but rather secondary immune thrombocytopenia³⁴ or secondary thrombotic microangiopathy, and is managed accordingly.

The platelet count thresholds for safe chemotherapy administration and the midcycle platelet count nadirs at which bleeding risk is increased are patient dependent. Patient-related risk factors include age, bleeding history, liver or kidney disease, presence of known bleeding disorders, concomitant antithrombotic therapy, and use of supplements and medications that may reduce platelet function. Cancer- and cancer treatment-specific factors include tumor type and location, tumor vascularity, radiation exposure, and use of agents that can result in endothelial damage (eg, cisplatin, antiangiogenic agents such as bevacizumab) or mucosal injury (eg, methotrexate, fluorouracil). Once these risks have been considered, platelet count thresholds of concern are assigned based on clinical judgement.

Ultimately, 2 subtypes of CIT are recognized in modern clinical practice, each of which has different implications for clinical management (Figure 1).⁴

Persistent CIT

Persistent CIT occurs when the platelet count is inadequate for on-time, full-dose administration of the prescribed chemotherapy regimen on day 1 of a chemotherapy cycle.^{4,27,35,36} In these patients, the platelet count on cycle day 1 is not usually low enough that the bleeding risk is substantially increased, but administration of full-dose chemotherapy at that time has an unacceptable risk of leading to a dangerously low and prolonged (several days or longer) midcycle count nadir resulting in high bleeding risk. In patients without excessive baseline bleeding risk, the platelet count threshold for safe administration of cytotoxic chemotherapy is usually adjudicated to be between $75 \times 10^9/L$ and $100 \times 10^9/L$, with a threshold near the lower end of this range for single-agent chemotherapy regimens and a threshold near the higher end for multiagent regimens. Persistent CIT most commonly presents in patients who have received numerous previous chemotherapy cycles and likely represents significant hematopoietic progenitor depletion and bone marrow microenvironment disruption from recurring cytotoxic insult.³⁷ For this reason, persistent CIT is usually chronic and recurrent even despite chemotherapy RDI reductions. This typically results in repeated treatment delays and compounding RDI reductions as additional cycles are given and the underlying hematopoietic compromise worsens. Therefore, persistent CIT is the type of CIT of greatest clinical relevance and the type most appropriate to treat with TPO-RA support in patients with solid tumors.^{30,38}

Nadir CIT

In contrast to persistent CIT, nadir CIT is characterized by normal or near-normal platelet counts on chemotherapy cycle day 1, with a severe or greater-than-expected drop in the platelet count at the time of the midcycle nadir.^{4,35,36} These

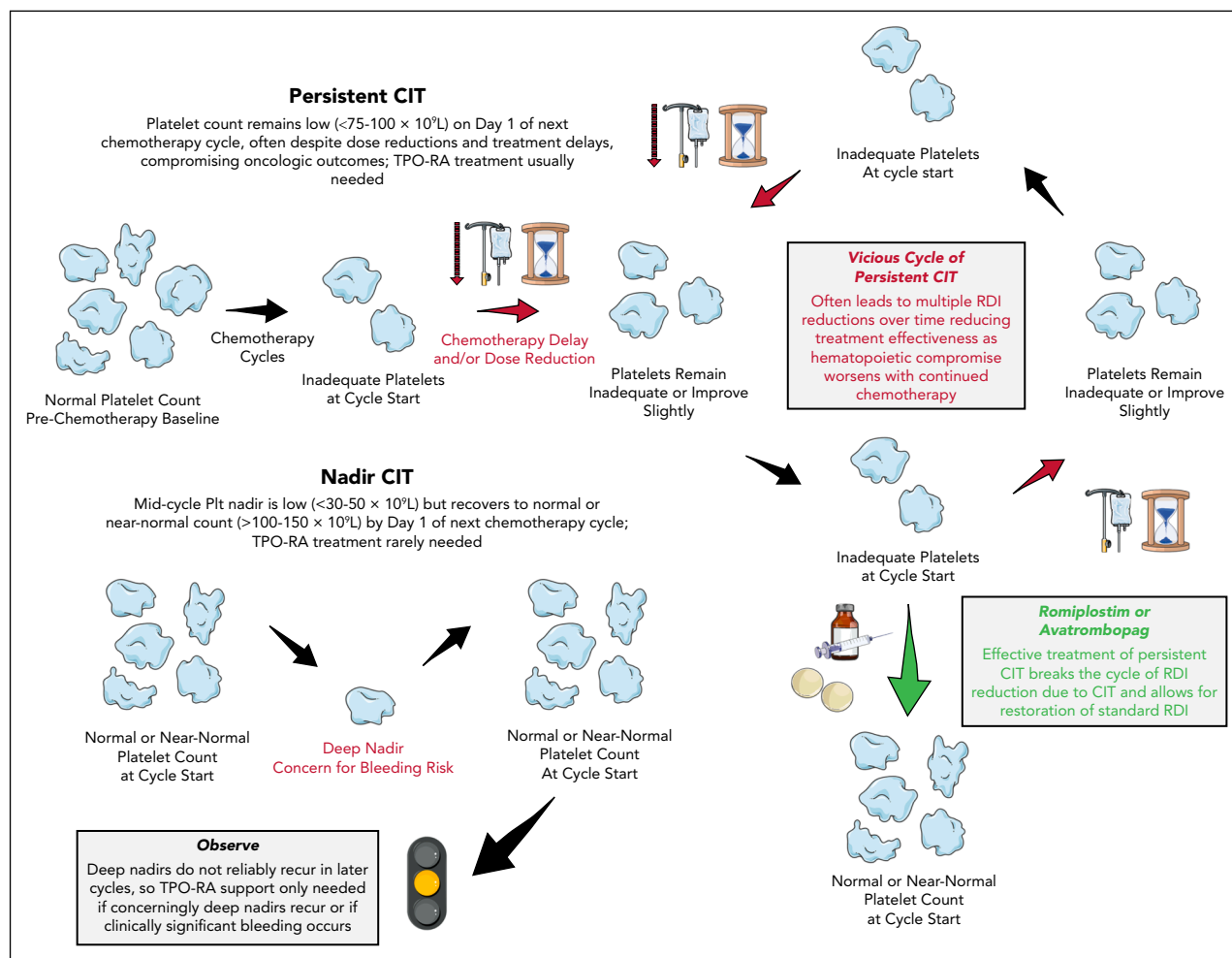


Figure 1. Summary of persistent CIT and nadir CIT. Graphics were reproduced and modified from Servier Medical Art (<https://smart.servier.com/>).

unusually deep nadirs are generally short-lived, typically resolving within 24 to 72 hours.³² As in patient case 1, nadir CIT in patients with solid tumors is usually only recognized if the patient has a platelet count measured midcycle during a toxicity check or develops bleeding complications from an excessive platelet count drop. The threshold for a concerning platelet nadir depends on the patient's bleeding risk: for patients with no history of bleeding from CIT and without excessive baseline bleeding risk, it is usually $<30 \times 10^9/L$ if the patient is not receiving antithrombotic therapy, and $<50 \times 10^9/L$ if the patient is receiving antithrombotic therapy (particularly therapeutic anticoagulation). Nadir CIT does not recur consistently with subsequent cycles of the same regimen even when the dose is not modified, particularly in patients with disease that is not heavily pretreated.³² For this reason, unless a patient has developed significant bleeding concurrently with the emergence of the nadir CIT, monitoring of subsequent cycle nadirs is indicated to confirm the concerning deep platelet count nadirs are recurrent before consideration of TPO-RA support.^{38,39} In addition, given the pharmacodynamics of TPO-RAs, initiation of the drug would need to be done at least 1 week before the cycle nadir (initiation at the time of the nadir is not likely to affect its depth or duration in a meaningful way). The patient in case 1, who has minor, nonspontaneous bleeding in the setting of nadir CIT, has no current indication for TPO-RA

support despite a single platelet count measured at, or around, $50 \times 10^9/L$ while on anticoagulation. In this case, monitoring platelet nadirs during the following 1 or 2 cycles to assess for recurrence of concerning nadirs and advising discontinuation of supplements that reduce platelet function (turmeric and fish oil) is most appropriate.

Role of platelet transfusion and antifibrinolytic therapy in CIT management

Transfused platelets in patients with cancer usually last for up to 3 days.⁴⁰ Platelet transfusions carry the risk of transfusion reactions, including a risk of sepsis higher than other blood products, can provoke platelet alloimmunization, and are a scarce resource.⁴⁰ In the overwhelming majority of cases, there is no role for routine prophylactic platelet transfusion in the management of CIT. The singular clear exception is the case of profound CIT ($<10 \times 10^9/L$ to $20 \times 10^9/L$) in patients without bleeding who are clearly at high risk of bleeding. Therapeutic platelet transfusion should generally be administered in patients with significant CIT ($<50 \times 10^9/L$) presenting with recent or ongoing major bleeding.

Although multiple randomized trials have found no significant benefit of antifibrinolytic therapy in patients with hematologic malignancies undergoing intensive chemotherapy and profound thrombocytopenia,^{41,42} no high-quality data exist for the use of these agents in patients with solid tumors experiencing bleeding in the setting of CIT. However, given the low risk of harm, adjunct use of antifibrinolytic therapy (eg, oral tranexamic acid 1000-1500 mg 2-3 times daily or epsilon-aminocaproic acid 1000 mg 3-4 times daily for a 3- to 5-day course) is reasonable for minor or clinically relevant nonmajor bleeding in the setting of CIT. Similarly, in patients presenting with major bleeding, IV administration of a 5000 mg tranexamic acid or epsilon-aminocaproic acid bolus in the acute emergent setting with subsequent infusion of 1000 mg hourly until hemostasis is achieved is reasonable.

Thrombopoietin receptor agonists for CIT in patients with solid tumors

CASE 2: PERSISTENT CIT

A 51-year-old man with oligometastatic colorectal adenocarcinoma receiving adjuvant fluorouracil, oxaliplatin, irinotecan, and leucovorin combination chemotherapy after colectomy and before planned curative resection of a solitary liver metastasis presents for administration of cycle 7, day 1 of chemotherapy with a platelet count of $74 \times 10^9/L$; the remainder of his complete blood count is unremarkable. Chemotherapy is delayed for 1 week due to thrombocytopenia. He presents the following week with a platelet count of $83 \times 10^9/L$, and dose-reduced chemotherapy is administered. Two weeks later, he presents for cycle 8, day 1 with a platelet count of $71 \times 10^9/L$. His current oncologic treatment plan is for 12 cycles of the current intensive regimen before liver resection. He is very anxious about the impact of thrombocytopenia on his ability to receive his chemotherapy as indicated and his chance for cancer cure.

Patient selection for thrombopoietin receptor agonist use in CIT

As described previously, CIT subtype (persistent or nadir) is the initial consideration in patient selection when considering use of TPO-RA support. In general, TPO-RA support is usually necessary to maintain RDI while also minimizing bleeding risk in persistent CIT but is usually unnecessary in nadir CIT, excepting those with recurring severe nadir CIT or a history of bleeding with nadir CIT. The patient in case 2 has classic persistent CIT, which has recurred and appears chronic despite both chemotherapy dose reduction and treatment delay. Initiation of TPO-RA support in this patient is likely to not only avert further chemotherapy dose reductions and treatment delays due to CIT but will also allow for dose escalation back to standard RDI given the previous dose reduction. Although the impact of CIT on RDI is especially acute in this case given the goal of cancer cure, the negative impact of RDI reductions on overall survival in patients treated with palliative intent would make TPO-RA support equally appropriate even if this patient was not curable.¹⁵⁻²¹ Indeed, most patients enrolled in studies of TPO-RA support of persistent CIT were treated for palliative intent for metastatic disease.^{25,30}

Although the published phase 3 trials of romiplostim and avatrombopag in CIT limited enrollment to patients with certain solid tumor types (3 different types in each trial per regulatory agency requests to minimize trial patient heterogeneity), the phase 2 studies and the large body of observational and real-world evidence that support these phase 3 trials include patients agnostic of tumor type and demonstrate effectiveness irrespective of tumor type.^{25,27,33} Therefore, we do not limit TPO-RA use to certain solid tumor types because there is no biological rationale to do so.

Aside from known hypersensitivity reaction or a previous serious adverse event directly attributable to TPO-RA use, there are few absolute contraindications to TPO-RA treatment of CIT. A history of venous or arterial thromboembolism is not a contraindication to TPO-RA support when otherwise indicated, particularly if the patient has completed the prescribed course of antithrombotic therapy or if it is ongoing at the time of TPO-RA initiation. This includes recently diagnosed thromboembolic events if they are appropriately managed with antithrombotic therapy. In fact, TPO-RA support in such patients with CIT will likely facilitate antithrombotic treatment that could otherwise be repeatedly interrupted due to thrombocytopenia. No randomized, controlled trial of TPO-RA treatment of CIT has shown a signal for higher rates of thromboembolic complications in patients with cancer receiving TPO-RAs compared with placebo, and rates of thromboembolic complications in observational studies are consistent with those of historical controls not receiving TPO-RA support.^{25-27,30,32} In addition, these agents do not cause platelet hyperreactivity, increased platelet activation, or spontaneous platelet aggregation,⁴³⁻⁴⁵ and thrombocytosis resulting from thrombopoietic agents has only been convincingly associated with thrombosis when extreme (ie, platelet counts well over $1000 \times 10^9/L$).^{46,47} It is reasonable to avoid, or use with caution, TPO-RAs in patients with a strong history of hypercoagulability (eg, triple-positive antiphospholipid antibody syndrome with multiple previous thromboembolic events) if they are not receiving indicated antithrombotic therapy to prevent further thrombosis. Beyond theoretical thromboembolic risk, other side-effects of romiplostim and avatrombopag are generally mild and uncommon in clinical practice, with mild to moderate headache (ie, usually readily managed with acetaminophen) being the most common.⁴⁸

Patient-specific predictors of nonresponse to TPO-RA treatment of CIT have been recognized.^{27,49} Although not precluding their use, these predictors of nonresponse complicate their use and require special considerations, discussed further in a subsequent section.

Agent selection and dosing of thrombopoietin receptor agonists in CIT

The agent with the most data supporting its use to treat CIT, and the only agent with positive phase 3 data, is the weekly subcutaneously administered romiplostim. Given this, guidance from both the National Comprehensive Cancer Network and the International Society on Thrombosis and Haemostasis recommends only romiplostim for routine off-label use.²⁷ Given the available data at the time of writing, romiplostim

should be considered the “default” agent of choice for off-label use. However, emerging data with avatrombopag, a second-generation oral TPO-RA with the benefit of oral administration and without dietary restrictions or hepatotoxicity, suggest that this agent may also be considered for off-label treatment of CIT.^{32,33} The only trials investigating eltrombopag in CIT evaluated its use for CIT prevention in unselected patients with cancer,⁵⁰⁻⁵² an approach that has been abandoned given the exposure of most such patients to unnecessary TPO-RA treatment. Moreover, the results of these studies raised questions of the overall efficacy of eltrombopag in patients receiving chemotherapy, given that the largest study did not show a significant improvement in platelet count between eltrombopag- and placebo-treated patients.⁵⁰ Data supporting potential use of eltrombopag in CIT treatment are very limited, and given efficacy concerns, known hepatotoxicity and required dietary restrictions in patients with cancer receiving emetogenic chemotherapy, we do not recommend use of eltrombopag to treat CIT. A summary of major clinical studies of romiplostim, eltrombopag, and avatrombopag in CIT is given in Table 1. Recombinant human thrombopoietin and hetrombopag are thrombopoietic agents currently available only in China and therefore will not be discussed further in this article.

Romiplostim Data for romiplostim include positive randomized phase 2 and 3 trials, headlined by the pivotal phase 3 RECITE trial, and a large body of high-quality observational real-world data.^{25-27,30} Figure 2 illustrates the authors’ recommended approach to initial dose selection, dose titration, and discontinuation for both futility and chemotherapy completion based on the published data and the authors’ experience. With adequate dosing, >80% of patients with persistent CIT respond well to romiplostim, attain adequate platelet counts, and maintain these counts durably over the course of continued chemotherapy so long as romiplostim support is continued.^{25-27,30} Clear loss of response should raise concerns for other complicating etiologies of thrombocytopenia, such as new bone marrow infiltration by tumor, infection, malignancy-associated thrombotic microangiopathy, or evolving chemotherapy-related clonal myeloid neoplasm. We continue romiplostim in patients experiencing cancer-associated thromboembolism while receiving it, so long as continued chemotherapy is planned, and adequate platelet counts provided by romiplostim facilitate indicated antithrombotic therapy.

As illustrated in Figure 2, we dose romiplostim weekly in CIT (this includes both persistent CIT and when indicated in nadir CIT) as is supported by published data. Although there are also observational data describing “intracycle” romiplostim dosing (administration of romiplostim on chemotherapy off-weeks only, or approximately every other week in patients receiving continuous chemotherapy), this approach results in higher bleeding rates and more RDI reductions.²⁷ Notably, thrombocytosis with weekly romiplostim is quite rare, with no thrombocytosis adverse events reported in the phase 3 RECITE trial.³⁰

Avatrombopag Data for avatrombopag in CIT are less established but include promising safety data and evidence of platelet count improvements from the global, pivotal phase 3

AVA-CIT-330 trial,³² which failed to meet its primary end point; as well as the multicenter US phase 2 randomized, controlled ACT-GI trial,³³ which did meet its primary end point of successful CIT treatment and recurrence prevention and provided additional evidence of safety of avatrombopag use in patients with cancer.

The phase 3 AVA-CIT-330 study was designed to evaluate efficacy of avatrombopag in patients with nadir CIT, and due to regulatory agency requests, limited the study population to relatively chemotherapy-naïve patients who had never had a previous history of CIT. Although the avatrombopag group had higher platelet counts, there was no difference in chemotherapy modifications due to CIT (the primary end point) between the avatrombopag and placebo groups due to lack of recurrence of nadir CIT in the placebo group.³² In short, the study evaluated the wrong patient population.⁵³ The subsequent ACT-GI study evaluated avatrombopag in patients with gastrointestinal cancers and persistent CIT, and was a positive study.³³ It is hoped that these findings will help establish avatrombopag as an option for persistent CIT and result in updates to international guidelines.

As with romiplostim, response to avatrombopag is generally durable and loss of response should raise concerns for other causes of thrombocytopenia. We continue avatrombopag in those experiencing cancer-associated thromboembolism while receiving it to facilitate ongoing chemotherapy and antithrombotic therapy unimpeded by fluctuating platelet counts. As illustrated in Figure 2, we dose avatrombopag daily in CIT with dose adjustment per platelet counts according to the ACT-GI protocol, not as a fixed dose for 5 days before and after chemotherapy administration, as was done in AVA-CIT-330.

CASE 3: COMPLICATED PERSISTENT CIT

A 45-year-old woman with triple-negative breast cancer metastatic to the lungs receiving fifth-line single-agent eribulin chemotherapy presents for cycle 4, day 1 of treatment. Treatment is delayed 1 week due to a platelet count of $47 \times 10^9/L$. Review of her platelet count trends show a significant drop upon switching from capecitabine to eribulin 2 months previous (attributed to the eribulin) and prompting chemotherapy dose reduction at cycle 2, but also a subsequent gradual decline in the platelet count measured on each cycle day 1 since that time. Further history reveals the onset of worsening hip pain over this time and abdominopelvic computed tomography scan demonstrates multiple bony metastases to the pelvis, lumbar spine, and right femur. Her peripheral blood film demonstrates occasional nucleated red cells, tear drop red cells, myelocytes, and metamyelocytes. Romiplostim $2 \mu g/kg$ per week is initiated to treat persistent CIT. However, the following week, her platelet count remains inadequate for treatment at $55 \times 10^9/L$.

Predictors of nonresponse to TPO-RAs in CIT

Predictors of nonresponse have been observed in patients receiving romiplostim to treat CIT.²⁷ These include bone marrow invasion by tumor, previous pelvic irradiation, and previous receipt of temozolomide, an alkylating agent with a distinct propensity for chronic myelosuppressive effects.^{54,55}

Table 1. Findings of major studies of TPO-RAs for the treatment of CIT in patients with solid tumors

Study Citation and aim	Patient population	Study design and identification	Chemotherapy regimen	Principal findings
Romiplostim Soff 2019 ²⁵ CIT treatment	60 patients with various solid tumors who developed persistent CIT (Plt of $<100 \times 10^9/L$ for ≥ 4 wks without chemotherapy treatment) randomized 2:1 to romiplostim or untreated observation; ultimately 52 received romiplostim	Randomized phase 2 trial (US, single-center); ClinicalTrials.gov identifier: NCT02052882	Various regimens	85% of patients treated with romiplostim achieved Plt count of $>100 \times 10^9/L$ compared with 12.5% randomized to untreated observation Only 7% of patients who achieved Plt count of $>100 \times 10^9/L$ experienced recurrent chemotherapy dose reduction or treatment delay 10.2% of patients had VTE over 12 months; bleeding not reported Subsequent publication ²⁶ confirmed similar safety and efficacy with extended duration use in the study population
Al-Samkari 2021 ²⁷ CIT treatment	173 patients with various malignancies (153 solid tumor, 20 lymphoma/myeloma) who developed persistent CIT (Plt count of $<100 \times 10^9/L$ for ≥ 3 wks or chemotherapy delay of ≥ 1 week due to thrombocytopenia)	Observational cohort (US, multicenter)	Various regimens	85% achieved Plt count of $>100 \times 10^9/L$ (95% without predictors of nonresponse) 71% achieved Plt count of $>75 \times 10^9/L$ and $\geq 30 \times 10^9/L$ higher than pretreatment baseline (82% without predictors of nonresponse) 79% avoided chemotherapy dose reduction or treatment delay; 89% avoided platelet transfusion Bone marrow tumor invasion, previous pelvic irradiation, previous temozolomide predicted romiplostim nonresponse VTE rate 14 per 100 patient-years; bleeding rate 23 per 100 patient-years (1% of 1063 cycles supported with romiplostim)
Al-Samkari 2026 ³⁰ CIT treatment	165 patients with various solid tumors who developed persistent CIT (Plt count of $<85 \times 10^9/L$) randomized 2:1 to romiplostim or placebo	Randomized phase 3 RECITE trial (study of romiplostim for chemotherapy-induced thrombocytopenia in adult patients with gastrointestinal, pancreatic or colorectal cancer) (global, multicenter); ClinicalTrials.gov identifier: NCT03362177	Oxaliplatin-based multiagent regimens	84% of patients treated with romiplostim had no CIT-induced modifications of chemotherapy dose compared with 36% treated with placebo Median platelet count nadir, likelihood of platelet response, and time to platelet response significantly higher in romiplostim group Duration-adjusted rate of grade ≥ 2 bleeding events 4.0 per 100 patient-years in romiplostim group and 7.6 per 100 patient-years in placebo group (did not achieve statistical significance) Thromboembolic events occurred in 2% in the romiplostim group, 0% in the placebo group No signal for increased risk of myelodysplastic syndrome or secondary malignancy in the romiplostim group over long-term follow-up

DSMB, data and safety monitoring board; Plt, platelet; US, United States; VTE, venous thromboembolism.

Table 1 (continued)

Study Citation and aim	Patient population	Study design and identification	Chemotherapy regimen	Principal findings
Avatrombopag				
Al-Samkari 2022 ²⁷ CIT treatment	122 patients with lung, ovarian, or bladder cancer with primarily nadir CIT randomized to receive avatrombopag 40 mg daily or placebo for 1 chemotherapy cycle	Randomized phase 3 AVA-CIT-330 trial (global, multicenter); ClinicalTrials.gov identifier: NCT03471078	Various regimens	Similar proportions of patients achieved the primary end point (of avoidance of chemotherapy treatment delay, dose reduction, bleeding, or platelet transfusion) in avatrombopag (70%) and placebo (73%) groups, due to high rates of unexpected higher platelet count nadirs in placebo group during the interventional chemotherapy cycle Avatrombopag-treated patients had higher platelet counts Avatrombopag overall safe and well tolerated in patients with cancer with a safety profile similar to that of placebo Arterial thromboembolism rate 2.4% in avatrombopag group vs 2.5% in placebo group No venous thromboembolic events in either group
Al-Samkari 2026 ³³ CIT treatment	47 patients with gastrointestinal cancers who developed persistent CIT (Plt count of $<85 \times 10^9/L$) randomized 1:1 to avatrombopag or placebo	Randomized phase 2 ACT-GI trial avatrombopag for chemotherapy-induced thrombocytopenia in gastrointestinal cancers (US, multicenter); ClinicalTrials.gov identifier: NCT05772546	Various regimens	Trial closed to enrollment by the DSMB at the planned interim analysis after meeting prespecified stopping criteria for efficacy 70% of patients treated with romiplostim, vs 17% treated with placebo, achieved the primary end point of correction of CIT and recurrence prevention (defined as recovery of platelets to $\geq 100 \times 10^9/L$, no chemotherapy dose modification/delay, and prevention of CIT recurrence [platelets $\geq 100 \times 10^9/L$] at end of the on-study cycle) Avatrombopag overall safe and well-tolerated with no treatment-related serious adverse events; no venous thromboembolic events in either group
Eltrombopag				
Kellum 2010 ⁵⁰ CIT prevention	183 chemotherapy-naïve patients with various advanced solid tumors randomized to placebo or eltrombopag 50, 75, or 100 mg; 134 patients completed at least 2 cycles and could be evaluated	Randomized phase 2 trial (international, multicenter); ClinicalTrials.gov identifier: NCT00102726	Carboplatin and paclitaxel	Primary end point (significant difference in the change in platelet count from day 1 in cycle 2 to the platelet nadir in cycle 2 between eltrombopag- and placebo-treated patients) not met Eltrombopag-treated patients had higher platelet counts at start of subsequent treatment cycles (higher counts with higher eltrombopag dose)
Winer 2017 ⁵² CIT prevention	75 patients with various solid tumors randomized to receive eltrombopag 100 mg or placebo; only 26 of the enrolled patients completed the planned number of cycles	Randomized phase 2 trial (international, multicenter); ClinicalTrials.gov identifier: NCT01147809	Gemcitabine or gemcitabine plus cisplatin or carboplatin	Eltrombopag-treated patients had higher platelet counts, lower frequencies of grade 3 or 4 CIT, more rapid platelet count recovery, and fewer dose reductions/treatment delays or missed doses due to thrombocytopenia Rates of grade 3/4 CIT remained high overall in both arms, raising concerns about potency of this agent in CIT

DSMB, data and safety monitoring board; Plt, platelet; US, United States; VTE, venous thromboembolism.

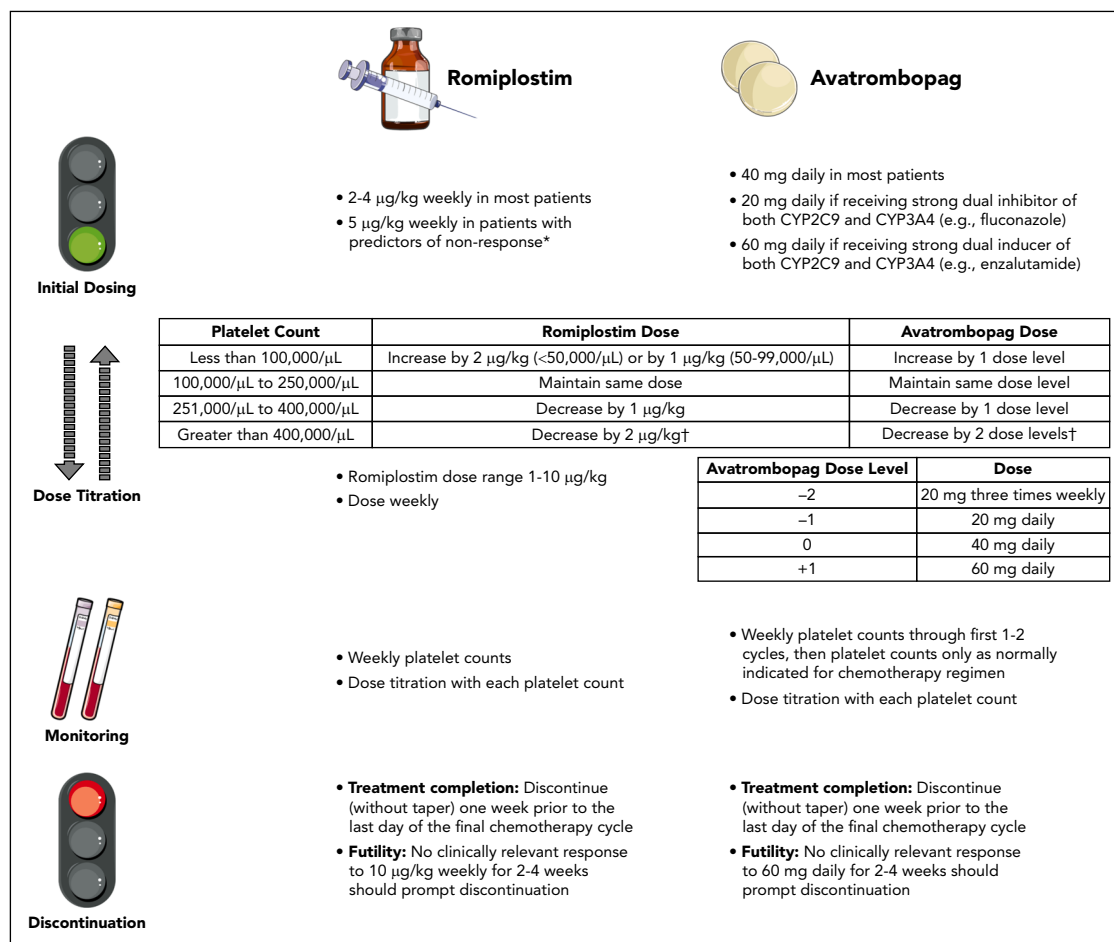


Figure 2. Dosing, titrating, monitoring, and discontinuing romiplostim and avatrombopag for CIT. Figure illustrates the authors' recommended approach to initial dose selection, dose titration, monitoring, and discontinuation for both futility and chemotherapy completion based on the published studies and the authors' experience. *Includes bone marrow invasion by tumor, previous pelvic irradiation (≥ 20 Gy), previous temozolomide, or drastic (>10 -20 times the upper limit of normal) endogenous TPO elevation. †A 1-week hold can be considered for profound thrombocytosis ($>1000 \times 10^9/\text{L}$) but is very rarely necessary. Graphics were reproduced and modified from Servier Medical Art (<https://smart.servier.com/>).

The common thread between these predictors of nonresponse is, unsurprisingly, compromise of the pelvic bone marrow, the primary source of hematopoietic tissue in adults. Patient case 3 demonstrates a patient with persistent CIT complicated by likely bone marrow tumor invasion. Although a bone marrow biopsy has not been performed to confirm marrow invasion, the presence of multiple bony metastatic lesions and leukoerythroblastosis on peripheral blood film in the setting of a tumor with a high propensity for marrow invasion is adequate for a presumptive diagnosis, and subjecting the patient to a bone marrow biopsy (which lacks perfect sensitivity for detection of invading tumor anyway) is unnecessary.

The presence of ≥ 1 predictors of nonresponse does not mean TPO-RA support is futile (Table 2); it means that adequate response is less likely and often requires higher TPO-RA dosing to achieve, and that magnitude of response may be blunted. We therefore initiate romiplostim at 4 to 5 $\mu\text{g/kg}$ in such patients, to improve the likelihood of a rapid response, and do not hesitate to uptitrate by 2 to 3 $\mu\text{g/kg}$ per week to achieve an adequate platelet count. Those not responding even at maximal doses of 10 $\mu\text{g/kg}$ per week should discontinue romiplostim for futility (Figure 2). Given the lack of data for avatrombopag in patients

with predictors of nonresponse, the greater dose titration potential and higher anticipated potency of romiplostim, we do not recommend avatrombopag in this setting.

Additional frequently asked questions relating to the use of TPO-RAs in the management of CIT, and their answers given the current published data, are listed in Table 3.

Thrombopoietin receptor agonists for CIT in patients with hematologic malignancies

CASE 4: ACUTE MYELOID LEUKEMIA

A 56-year-old man with recently diagnosed acute myeloid leukemia develops severe thrombocytopenia after receipt of induction chemotherapy. His platelet count is $15 \times 10^9/\text{L}$ and he is receiving platelet transfusions for a platelet count $<10 \times 10^9/\text{L}$. He does not have bleeding currently. The question of TPO-RA support to minimize platelet transfusions and hasten platelet count recovery is raised.

Table 2. Patient, CIT, and cancer-related factors influencing candidacy for TPO-RA treatment of CIT

Ideal	Suboptimal but not contraindicated	Inappropriate
Persistent CIT	Nadir CIT with previous bleeding or recurrent concerning low platelet nadirs	High-grade myeloid malignancy (eg, high-grade myelodysplastic syndrome, acute myeloid leukemia)
Solid tumor malignancy	Bone marrow infiltration by tumor (predictor of nonresponse) ²⁷	
Normal or only modest elevation (<5 times upper limit of normal) of endogenous thrombopoietin ⁴⁹	Previous pelvic irradiation (predictor of nonresponse) ²⁷	
No evidence of bone marrow failure (adequate absolute neutrophil count [$>1.0 \times 10^9/L$] and hemoglobin [$>7-8$ g/dL])	Previous temozolomide treatment (predictor of nonresponse) ²⁷	
	Diffuse bone metastases	
	Lymphoid malignancy ^{27,56,57}	
	Dramatic elevations of endogenous thrombopoietin ($>10-20$ times upper limit of normal) ⁴⁹	

Table 3. Additional frequently asked questions pertaining to TPO-RA use in CIT

Frequently Asked Question	Answer
Administration timing relative to chemotherapy Does the administration of a TPO-RA need to be avoided on chemotherapy infusion days due to concern for bone marrow sensitization, as is done with G-CSF?	No. Thrombopoietic agents do not sensitize bone marrow progenitors to chemotherapy and can be administered on the day of chemotherapy infusion.
Combination with granulocyte colony-stimulating factors Can TPO-RAs be combined with granulocyte colony-stimulating factors? If so, should one be concerned for worsened toxicity of stimulating hematopoiesis (eg, bone pain or splenic rupture)?	Yes. TPO-RAs can be combined safely with G-CSFs, and no, there are no data suggesting a concern for worsening bone pain or increased risk of splenic rupture when doing so.
Use of TPO-RAs in chemotherapy-naïve patients to prevent CIT Is there any role in using TPO-RAs in patients with normal platelet counts before chemotherapy initiation to prevent the development of CIT in the first place?	No. Previous studies with eltrombopag in CIT interrogated this question. This results in unnecessary TPO-RA use in most such patients.
Use of TPO-RAs in pediatric patients with CIT Can TPO-RAs be used in pediatric patients with solid tumors and CIT?	Yes. Although the data are limited to smaller observational studies, ^{58,59} this early data suggest it is safe and effective to use romiplostim to treat CIT in pediatric patients. Recognizing data are limited, there is no biological rationale why the principles established by the adult data would not apply to children with significant persistent CIT and clinical need.
Rebound thrombocytopenia after discontinuation/need to taper Rebound thrombocytopenia can be seen with abrupt TPO-RA discontinuation in immune thrombocytopenia, particularly with romiplostim. With use of these agents in CIT, is rebound thrombocytopenia a concern, and is tapering necessary?	No. Rebound thrombocytopenia has not been described with any TPO-RA in CIT, and tapering is not necessary at the time of discontinuation.
Need to assess for thrombophilia Given the theoretical risk of thromboembolism with TPO-RAs and the increased risk of thromboembolism in patients with cancer, is thrombophilia evaluation, including antiphospholipid antibody testing and hereditary thrombophilia testing, indicated before starting a TPO-RA for CIT?	No. There is no evidence that this is necessary and we believe it to be inappropriate excessive testing. Clinical trials of TPO-RAs in CIT did not do this. Similarly, this would be inappropriate in any other situation in which TPO-RAs are used (eg, immune thrombocytopenia, aplastic anemia, etc.)
Need to initiate antithrombotic prophylaxis Given the theoretical risk of thromboembolism with TPO-RAs, is initiation of thromboprophylaxis with aspirin or prophylactic-dose anticoagulation indicated when initiating these agents in cancer patients with tumor types at high risk for thrombosis, such as pancreatic or brain cancers?	No. No study suggests the need to do this, and data regarding thromboembolism risk of TPO-RAs in CIT have been quite favorable in every randomized trial irrespective of agent. We do not consider TPO-RA use when making decisions about the use of primary prophylaxis of cancer-associated thrombosis in individual patients.

EPO, erythropoietin; G-CSF, granulocyte colony-stimulating factor; ISTH, International Society on Thrombosis and Haemostasis; NCCN, National Comprehensive Cancer Network.

Table 3 (continued)

Frequently Asked Question	Answer
Use of TPO-RAs to support antithrombotic therapy When a patient has a new requirement for antithrombotic therapy but their degree of CIT precludes its safe use, can TPO-RAs be used to facilitate antithrombotic therapy with ongoing chemotherapy?	Yes. As mentioned previously, there is no single platelet count threshold for CIT. When a patient initiates antithrombotic therapy, the risk-benefit calculus for proceeding with full-dose, on-time chemotherapy at low platelet counts may change. For that patient, if proper antithrombotic and antineoplastic therapy cannot proceed safely with the current platelet counts, it is reasonable to initiate TPO-RAs to facilitate this.
Use in lymphoma and plasma cell neoplasms Can TPO-RAs be used to treat CIT in patients with lymphoma or plasma cell neoplasms?	Yes. Albeit with much less supporting evidence than in solid tumors. ^{27,56,57} The likelihood of response appears lower and responses are often blunted, likely due to marrow involvement by these tumors. There are no special safety concerns with use of TPO-RAs in these patients, however.
Promotion of leukemogenesis or myelofibrosis Can TPO-RAs promote the development of myeloid neoplasms or leukemia, or myelofibrosis in patients with cancer?	No. High-quality data, including longer-term follow-up data, in CIT, immune thrombocytopenia, and myelodysplastic syndrome have alleviated historical theoretical concerns regarding the potential for leukemogenicity, promotion of hematopoietic clonal evolution, and development of irreversible myelofibrosis with the use of TPO-RAs. ^{30,60,61}
Stimulate tumor growth Because cancer cells can express the EPO receptor and erythropoiesis-stimulating agents can stimulate solid tumor growth, is this a concern with TPO-RAs and TPO receptor expression in solid tumors?	No. Nonmyeloid tumor cells do not exhibit meaningful expression of the thrombopoietin receptor ⁶² and thrombopoietin agonism does not stimulate non-myeloid tumor cell growth. ⁶³
Facilitate tumor metastasis Some data suggest that platelets can be hijacked by tumor cells to promote metastatic spread. ⁶⁴ Is this a concern with the use of TPO-RAs to raise the platelet count?	No. These preclinical data are important and merit further study, but there is no evidence from any TPO-RA clinical trial or TPO-RA clinical data in general that this is a concern with use of these agents in CIT. The goal with the use of TPO-RAs in CIT is maintaining a platelet count at the low end of the physiological range (not induction of supraphysiologic or even high normal platelet counts). Notably, and relevant to this question, thrombocytopenia has not been associated with reduced cancer metastasis or improved survival in patients with cancer; it is associated with a worse prognosis. ⁶⁵
Concern with concurrent radiation therapy Use of granulocyte colony-stimulating factors concurrent with radiation therapy to certain locations may increase the risk of radiation toxicity; eg, there is a known risk of severe lung toxicity and thrombocytopenia when these agents are used concurrently with thoracic radiation. Does this risk exist with use of TPO-RAs during radiation therapy?	No, to the best of our knowledge at this time, but this specific situation has not been rigorously evaluated in CIT studies.
Predictive value of endogenous TPO levels Can endogenous TPO levels be used to predict the likelihood of response to TPO-RAs in CIT, like use of endogenous EPO levels before starting erythropoiesis-stimulating agents?	Yes. Endogenous TPO levels have been observed to predict response to TPO-RAs across numerous conditions, including immune thrombocytopenia, CIT, and thrombocytopenia of chronic liver disease. ^{49,66-68} In one retrospective study of romiplostim in persistent CIT, dramatic elevations of the endogenous serum thrombopoietin level to greater than 10-20 times the upper limit of the normal range strongly predicted nonresponse to romiplostim, whereas those with normal or mild elevations in endogenous thrombopoietin (<5 times the upper limit of normal) were much more likely to respond to romiplostim. ⁶⁸ When turnaround time for TPO level testing is short, it is reasonable to consider this testing during initial CIT evaluation in any patient with a solid tumor and persistent CIT to help guide the use of romiplostim.
TPO-RA combination therapy and TPO-RA switching in CIT If a patient does not respond to one TPO-RA to treat CIT, are there data to support that they may respond to another, or might respond to combination TPO-RA therapy, as is the case in immune thrombocytopenia?	No. Although combination thrombopoietic therapy ⁶⁹ or switching from one TPO-RA to another ⁷⁰⁻⁷² has been successful in immune thrombocytopenia, the etiology of treatment failure in CIT is biologically different and data is lacking to support these approaches in this scenario.
Payer coverage and access to TPO-RAs in CIT Because these agents remain off-label, how does one obtain access to TPO-RAs to treat CIT?	Although we hope that the positive data for these agents will ultimately lead to regulatory approvals, at the time of writing all TPO-RAs remain off-label for CIT and this can be a barrier to access. The approach to accessing these drugs varies by country, but in the United States, access is strongly supported by citing the published studies listed in Table 1 , particularly for romiplostim, as well as the published guidelines from NCCN and ISTH recommending use of these agents. We recommend citing these studies and guidelines.

EPO, erythropoietin; G-CSF, granulocyte colony-stimulating factor; ISTH, International Society on Thrombosis and Haemostasis; NCCN, National Comprehensive Cancer Network.

To date, all the high-quality data demonstrating safety and efficacy of TPO-RAs in the management of CIT are in patients with solid tumors. Although the first-generation thrombopoietic agents and TPO-RAs have been evaluated to hasten platelet count recovery after induction or consolidation chemotherapy in patients with high-grade myeloid malignancies, these studies have been negative with no clear benefit of TPO-RA use.^{73,74} Moreover, unlike in solid tumors and lymphoid malignancies, myeloid cancer cells may express thrombopoietin receptors and use of these agents can stimulate tumor cell growth.^{75,76} Therefore, the use of TPO-RAs to treat CIT in patients with high-grade myeloid malignancies, as with the patient in case 4, is not recommended. TPO-RAs have a role in the treatment of thrombocytopenia in lower-grade myelodysplastic syndromes and have demonstrated safety and efficacy in this setting^{77,78}; however, this use is not specifically for the treatment of CIT and further discussion is out of scope for this article.

In patients undergoing allogeneic or autologous hematopoietic stem cell transplantation, existing studies have not demonstrated more rapid recovery of platelets or other cell counts with the use of TPO-RA support.⁵⁶ Studies are ongoing, but use of TPO-RAs in this setting is not currently recommended.

Use of TPO-RAs to treat CIT in patients with lymphoid malignancies is controversial, largely owing to a paucity of high-quality data. In cases in which TPO-RA use has been evaluated in this setting, it has been successful, albeit in a lower percentage of patients than in those with solid tumors, probably owing to the propensity of these tumors to invade the bone marrow.^{27,57} Therefore, we do use TPO-RA support for CIT in lymphoid malignancies according to the solid tumor rubric, when clearly necessary, with acknowledgment of, and disclosure to patients of, the limited supporting data.

Conclusions

Management of CIT is an essential component of oncologic care. The evidence pertaining to the use of the TPO-RA romiplostim in the management of persistent CIT in patients with solid tumors is now fully mature. It demonstrates that these agents are safe and efficacious in maintaining chemotherapy RDI and avoiding bleeding in this setting, and that the benefits are expected to considerably outweigh the risks. The

use of TPO-RAs can be considered in limited patients with nadir CIT, only once clinical need has been confirmed. Although supportive care to maintain RDI in other settings, such as chemotherapy-induced neutropenia, has been shown to improve survival, an important caveat of TPO-RA support of CIT is that it has yet to be demonstrated specifically in this setting (although studies are ongoing). Both romiplostim and avatrombopag have high-quality randomized, controlled trial-level evidence to support their use in persistent CIT, although the evidence base is much more complete for romiplostim. Proper use of TPO-RAs in CIT requires unique dosing and considerations unlike any other indications for these agents. However, all these precepts apply only to patients with solid tumors (and to a much lesser extent, lymphoid malignancies); there is no role for the use of TPO-RAs to manage CIT in patients with high-grade myeloid malignancies. It is hoped that the accumulated evidence for the use of TPO-RAs in CIT will ultimately lead to the regulatory approval of these agents for this indication.

Authorship

Contribution: H.A.-S. contributed to writing of the first draft of the manuscript, and creation of the tables and figures; and H.A.-S. and G.A.S. contributed to manuscript conceptualization and design, and critical revision of the manuscript and final approval.

Conflict-of-interest disclosure: H.A.-S. reports consultancy role with Agios, Amgen, Alnylam, Alpine, Atavistik, Diagonal, Novartis, Pharmacosmos, Sobi, Sanofi, Takeda, Terremoto, Tectonic, and Vaderis and reports research funding to the institution from Agios, Amgen, Alnylam, Novartis, Sobi, Atavistik, and Vaderis. G.A.S. reports grants from, or contracts with, Amgen, Sobi, and Anthos; reports participation on a data safety monitoring board for Alpine; and reports participation on advisory boards for Sanofi and Novartis.

ORCID profile: H.A.-S., [0000-0001-6175-1383](https://orcid.org/0000-0001-6175-1383).

Correspondence: Hanny Al-Samkari, Division of Classical Hematology, Harvard Medical School, Massachusetts General Hospital, 55 Fruit St, Bartlett Hall Extension Office 133, Boston, MA 02114; email: hal-samkari@mgh.harvard.edu.

Footnote

Submitted 6 April 2026; accepted 19 June 2026; prepublished online on *Blood* First Edition 6 July 2026. <https://doi.org/10.1182/blood.2026034099>.

REFERENCES

- Epstein RS, Aapro MS, Basu Roy UK, et al. Patient burden and real-world management of chemotherapy-induced myelosuppression: Results from an online survey of patients with solid tumors. *Adv Ther*. 2020;37(8):3606-3618.
- National Comprehensive Cancer Network. Hematopoietic Growth Factors. Accessed 15 March 2026. <https://www.nccn.org/guidelines/guidelines-detail?category=3&id=1493>
- Wu Y, Aravind S, Ranganathan G, Martin A, Nalysnyk L. Anemia and thrombocytopenia in patients undergoing chemotherapy for solid tumors: a descriptive study of a large outpatient oncology practice database, 2000-2007. *Clin Ther*. 2009;31(Pt 2):2416-2432.
- Al-Samkari H, Soff GA. Clinical challenges and promising therapies for chemotherapy-induced thrombocytopenia. *Expert Rev Hematol*. 2021;14(5):437-448.
- Kuter DJ. Treatment of chemotherapy-induced thrombocytopenia in patients with non-hematologic malignancies. *Haematologica*. 2022;107(6):1243-1263.
- Cheloff AZ, Al-Samkari H. Romiplostim for PARP inhibitor-induced thrombocytopenia in solid tumor malignancies. *Platelets*. 2022;33(8):1312-1313.
- Elting LS, Rubenstein EB, Martin CG, et al. Incidence, cost, and outcomes of bleeding and chemotherapy dose modification among solid tumor patients with chemotherapy-induced thrombocytopenia. *J Clin Oncol*. 2001;19(4):1137-1146.
- Stanworth SJ, Estcourt LJ, Powter G, et al. A no-prophylaxis platelet-transfusion strategy for hematologic cancers. *N Engl J Med*. 2013;368(19):1771-1780.
- Kayano SS, Santana PV, Colella R, Colella MP, Caruso P. Lower platelet count and metastatic tumor are associated with increased risk of spontaneous bleeding in critically ill patients with cancer: An observational study. *Transfusion*. 2023;63(12):2311-2320.

10. Patell R, Hsu C, Shi M, et al. Impact of mild thrombocytopenia on bleeding and recurrent thrombosis in cancer. *Haematologica*. 2024;109(6):1849-1856.
11. Al-Samkari H, Connors JM. Managing the competing risks of thrombosis, bleeding, and anticoagulation in patients with malignancy. *Blood Adv*. 2019;3(22):3770-3779.
12. Leiva O, Newcomb R, Connors JM, Al-Samkari H. Cancer and thrombosis: new insights to an old problem. *J Med Vasc*. 2020;45(6):6S8-6S16.
13. Alwhaibi M, AlRuthia Y, Alhawassi TM, Almalag H, Alsalloum H, Balkhi B. Polypharmacy and comorbidities among ambulatory cancer patients: a cross-sectional retrospective study. *J Oncol Pharm Pract*. 2020;26(5):1052-1059.
14. Lam CS, Hua R, Au-Doung LW, et al. Potential supplement-drug interactions and bleeding in patients with cancer: a UK Biobank study. *J Integr Med*. 2026;24(1):90-97.
15. Crawford J, Denduluri N, Patt D, et al. Relative dose intensity of first-line chemotherapy and overall survival in patients with advanced non-small-cell lung cancer. *Support Care Cancer*. 2020;28(2):925-932.
16. Lyman GH, Dale DC, Crawford J. Incidence and predictors of low dose-intensity in adjuvant breast cancer chemotherapy: a nationwide study of community practices. *J Clin Oncol*. 2003;21(24):4524-4531.
17. Lyman GH, Dale DC, Friedberg J, Crawford J, Fisher RI. Incidence and predictors of low chemotherapy dose-intensity in aggressive non-Hodgkin's lymphoma: a nationwide study. *J Clin Oncol*. 2004;22(21):4302-4311.
18. Claessens AKM, Bos M, Lopez-Yurda M, et al. Intermittent versus continuous first-line treatment for HER2-negative metastatic breast cancer: the Stop & Go study of the Dutch Breast Cancer Research Group (BOOG). *Breast Cancer Res Treat*. 2018;172(2):413-423.
19. Hanna RK, Poniewierski MS, Laskey RA, et al. Predictors of reduced relative dose intensity and its relationship to mortality in women receiving multi-agent chemotherapy for epithelial ovarian cancer. *Gynecol Oncol*. 2013;129(1):74-80.
20. Havrilesky LJ, Reiner M, Morrow PK, Watson H, Crawford J. A review of relative dose intensity and survival in patients with metastatic solid tumors. *Crit Rev Oncol Hematol*. 2015;93(3):203-210.
21. Nakayama G, Tanaka C, Uehara K, et al. The impact of dose/time modification in irinotecan- and oxaliplatin-based chemotherapies on outcomes in metastatic colorectal cancer. *Cancer Chemother Pharmacol*. 2014;73(4):847-855.
22. Nielson CM, Bylsma LC, Fryzek JP, Saad HA, Crawford J. Relative dose intensity of chemotherapy and survival in patients with advanced stage solid tumor cancer: a systematic review and meta-analysis. *Oncologist*. 2021;26(9):e1609-e1618.
23. Lyman GH, Dale DC, Culakova E, et al. The impact of the granulocyte colony-stimulating factor on chemotherapy dose intensity and cancer survival: a systematic review and meta-analysis of randomized controlled trials. *Ann Oncol*. 2013;24(10):2475-2484.
24. Lyman GH, Yau L, Nakov R, Krendyukov A. Overall survival and risk of second malignancies with cancer chemotherapy and G-CSF support. *Ann Oncol*. 2018;29(9):1903-1910.
25. Soff GA, Miao Y, Bendheim G, et al. Romiplostim treatment of chemotherapy-induced thrombocytopenia. *J Clin Oncol*. 2019;37(31):2892-2898.
26. Wilkins CR, Ortiz J, Gilbert LJ, et al. Romiplostim for chemotherapy-induced thrombocytopenia: efficacy and safety of extended use. *Res Pract Thromb Haemost*. 2022;6(3):e12701.
27. Al-Samkari H, Parnes AD, Goodarzi K, Weitzman JI, Connors JM, Kuter DJ. A multicenter study of romiplostim for chemotherapy-induced thrombocytopenia in solid tumors and hematologic malignancies. *Haematologica*. 2021;106(4):1148-1157.
28. Al-Samkari H, Marshall AL, Goodarzi K, Kuter DJ. The use of romiplostim in treating chemotherapy-induced thrombocytopenia in patients with solid tumors. *Haematologica*. 2018;103(4):e169-e172.
29. Soff GA, Al-Samkari H, Leader A, Eisen M, Saad H. Romiplostim in chemotherapy-induced thrombocytopenia: a review of the literature. *Cancer Med*. 2024;13(15):e7429.
30. Al-Samkari H, Munoz C, Geredeli C, et al. Romiplostim versus placebo for chemotherapy-induced thrombocytopenia. *N Engl J Med*. 2026;394(11):1061-1073.
31. Soff G, Leader A, Al-Samkari H, et al. Management of chemotherapy-induced thrombocytopenia: guidance from the ISTH subcommittee on hemostasis and malignancy. *J Thromb Haemost*. 2024;22(1):53-60.
32. Al-Samkari H, Kolb-Sielecki J, Safina SZ, Xue X, Jamieson BD. Avatrombopag for chemotherapy-induced thrombocytopenia in patients with non-haematological malignancies: an international, randomised, double-blind, placebo-controlled, phase 3 trial. *Lancet Haematol*. 2022;9(3):e179-e189.
33. Al-Samkari H, Shatzel JJ, Panch SR, et al. Avatrombopag versus placebo for persistent chemotherapy-induced thrombocytopenia in gastrointestinal cancers: the phase II ACT-GI trial. *J Clin Oncol*. Published online 13 July 2026. doi.org/10.1200/JCO-26-00568
34. Leaf RK, Mones JV, Shenoy T, et al. Immune thrombocytopenia in patients treated with immune checkpoint inhibitors. *Blood*. 2026;147(12):1351-1364.
35. Al-Samkari H. Thrombopoietin receptor agonists for chemotherapy-induced thrombocytopenia: a new solution for an old problem. *Hematol Am Soc Hematol Educ Program*. 2022;2022(1):286-295.
36. Al-Samkari H. Optimal management of chemotherapy-induced thrombocytopenia with thrombopoietin receptor agonists. *Blood Rev*. 2024;63:101139.
37. Peshin S, Dharia A, Takrori E, Kaur J, Thanikachalam K, Iyer R. Understanding chemotherapy-induced thrombocytopenia: implications for gastrointestinal cancer treatment. *Curr Oncol*. 2025;32(8):455.
38. Al-Samkari H. Should thrombopoietin receptor agonists be used for chemotherapy-induced thrombocytopenia? *Res Pract Thromb Haemost*. 2025;9(5):102980.
39. Song AB, Al-Samkari H. Chemotherapy-induced thrombocytopenia: modern diagnosis and treatment. *Br J Haematol*. 2025;206(4):1062-1066.
40. Schiffer CA, Bohlke K, Delaney M, et al. Platelet transfusion for patients with cancer: American Society of Clinical Oncology clinical practice guideline update. *J Clin Oncol*. 2018;36(3):283-299.
41. Gernsheimer TB, Brown SP, Triulzi DJ, et al. Prophylactic tranexamic acid in patients with hematologic malignancy: a placebo-controlled, randomized clinical trial. *Blood*. 2022;140(11):1254-1262.
42. Alsuliman T, Chalayer E, Stocker N, et al. Tranexamic acid to prevent bleeding in patients with hematologic malignancies and hypoproliferative thrombocytopenia: a systematic review and meta-analysis of randomized controlled trials with trial sequential analysis. *J Thromb Haemost*. 2026;24(1):99-107.
43. Michelson AD, Smolensky Koganov E, Forde EE, Carmichael SL, Frelinger AL 3rd. Avatrombopag increases platelet count but not platelet activation in patients with thrombocytopenia resulting from liver disease. *J Thromb Haemost*. 2018;16(12):2515-2519.
44. Psaila B, Bussell JB, Linden MD, et al. In vivo effects of eltrombopag on platelet function in immune thrombocytopenia: no evidence of platelet activation. *Blood*. 2012;119(17):4066-4072.
45. Al-Samkari H, Van Cott EM, Kuter DJ. Platelet aggregation response in immune thrombocytopenia patients treated with romiplostim. *Ann Hematol*. 2019;98(3):581-588.
46. Harker LA, Marzec UM, Hunt P, et al. Dose-response effects of pegylated human megakaryocyte growth and development factor on platelet production and function in nonhuman primates. *Blood*. 1996;88(2):511-521.
47. Harker LA, Hunt P, Marzec UM, et al. Regulation of platelet production and function by megakaryocyte growth and development factor in nonhuman primates. *Blood*. 1996;87(5):1833-1844.

48. Al-Samkari H, Kuter DJ. Immune thrombocytopenia in adults: modern approaches to diagnosis and treatment. *Semin Thromb Hemost.* 2020;46(3):275-288.
49. Song AB, Goodarzi K, Karp Leaf R, Kuter DJ, Al-Samkari H. Thrombopoietin level predicts response to treatment with romiplostim in chemotherapy-induced thrombocytopenia. *Am J Hematol.* 2021;96(12):1563-1568.
50. Kellum A, Jagiello-Gruszfeld A, Bondarenko IN, Patwardhan R, Messam C, Mostafa Kamel Y. A randomized, double-blind, placebo-controlled, dose ranging study to assess the efficacy and safety of eltrombopag in patients receiving carboplatin/paclitaxel for advanced solid tumors. *Curr Med Res Opin.* 2010;26(10):2339-2346.
51. Winer ES, Safran H, Karaszewska B, et al. Eltrombopag with gemcitabine-based chemotherapy in patients with advanced solid tumors: a randomized phase I study. *Cancer Med.* 2015;4(1):16-26.
52. Winer ES, Safran H, Karaszewska B, et al. Eltrombopag for thrombocytopenia in patients with advanced solid tumors receiving gemcitabine-based chemotherapy: a randomized, placebo-controlled phase 2 study. *Int J Hematol.* 2017;106(6):765-776.
53. Al-Samkari H, Moore DC. A second chance for avatrombopag in chemotherapy-induced thrombocytopenia. *Br J Haematol.* 2025; 206(1):375-377.
54. Villano JL, Letarte N, Yu JM, Abdur S, Bressler LR. Hematologic adverse events associated with temozolomide. *Cancer Chemother Pharmacol.* 2012;69(1):107-113.
55. Park AK, Waheed A, Forst DA, Al-Samkari H. Characterization and prognosis of temozolomide-induced aplastic anemia in patients with central nervous system malignancies. *Neuro Oncol.* 2022;24(6): 964-973.
56. Scordo M, Gilbert LJ, Hanley DM, et al. Open-label pilot study of romiplostim for thrombocytopenia after autologous hematopoietic cell transplantation. *Blood Adv.* 2023;7(8):1536-1544.
57. Joffe E, Epstein-Peterson Z, Falchi L, et al. Romiplostim for prevention of severe chemotherapy-induced thrombocytopenia in lymphoma patients-phase I study. *Cancers (Basel).* 2026;18(2):188.
58. Degliuomini MD, Armstrong K, Curry M, et al. A single institutional experience using romiplostim to treat therapy-related thrombocytopenia in pediatric solid tumor patients. *Pediatr Blood Cancer.* 2025;72(11): e31977.
59. Gorfinkel L, Harris EM, Wall CB, O'Neill AF. Romiplostim for chemotherapy-induced thrombocytopenia: a case cohort assessment of safety, efficacy, and proposed future study in pediatric patients with solid tumors. *Pediatr Blood Cancer.* 2025;72(11):e32011.
60. Kantarjian HM, Fenaux P, Sekeres MA, et al. Long-term follow-up for up to 5 years on the risk of leukaemic progression in thrombocytopenic patients with lower-risk myelodysplastic syndromes treated with romiplostim or placebo in a randomised double-blind trial. *Lancet Haematol.* 2018; 5(3):e117-e126.
61. Kuter DJ, Bussel JB, Newland A, et al. Long-term treatment with romiplostim in patients with chronic immune thrombocytopenia: safety and efficacy. *Br J Haematol.* 2013; 161(3):411-423.
62. Erickson-Miller CL, Pillarisetti K, Kirchner J, et al. Low or undetectable TPO receptor expression in malignant tissue and cell lines derived from breast, lung, and ovarian tumors. *BMC Cancer.* 2012;12:405.
63. Erickson-Miller CL, Chadderton A, Gibbard A, et al. Thrombopoietin receptor levels in tumor cell lines and primary tumors. *J Oncol.* 2010;2010:135354.
64. Liao K, Zhang X, Liu J, et al. The role of platelets in the regulation of tumor growth and metastasis: the mechanisms and targeted therapy. *Med Comm (2020).* 2023; 4(5):e350.
65. Adelborg K, Veres K, Horvath-Puho E, Clouser M, Saad H, Sorensen HT. Risk and adverse clinical outcomes of thrombocytopenia among patients with solid tumors-a Danish population-based cohort study. *Br J Cancer.* 2024;130(9):1485-1492.
66. Al-Samkari H, Kuter DJ. Thrombopoietin level predicts response to treatment with eltrombopag and romiplostim in immune thrombocytopenia. *Am J Hematol.* 2018; 93(12):1501-1508.
67. Al-Samkari H. Avatrombopag maleate for the treatment of periprocedural thrombocytopenia in patients with chronic liver disease. *Drugs Today (Barc).* 2018; 54(11):647-655.
68. Makar RS, Zhukov OS, Sahud MA, Kuter DJ. Thrombopoietin levels in patients with disorders of platelet production: diagnostic potential and utility in predicting response to TPO receptor agonists. *Am J Hematol.* 2013; 88(12):1041-1044.
69. Park AK, Park JC, Al-Samkari H. Pembrolizumab-induced acquired megakaryocytic thrombocytopenia and successful combination treatment with eltrombopag, romiplostim and cyclosporine: a brief communication. *J Immunother.* 2022; 45(7):321-323.
70. Gonzalez-Porras JR, Godeau B, Carpenedo M. Switching thrombopoietin receptor agonist treatments in patients with primary immune thrombocytopenia. *Ther Adv Hematol.* 2019;10:2040620719837906.
71. Al-Samkari H, Jiang D, Gernsheimer T, et al. Adults with immune thrombocytopenia who switched to avatrombopag following prior treatment with eltrombopag or romiplostim: a multicentre US study. *Br J Haematol.* 2022; 197(3):359-366.
72. Al-Samkari H, Jiang D, Gernsheimer T, et al. Durability of platelet response after switching to avatrombopag from eltrombopag or romiplostim in immune thrombocytopenia. *Res Pract Thromb Haemost.* 2023;7(3):100134.
73. Schiffer CA, Miller K, Larson RA, et al. A double-blind, placebo-controlled trial of pegylated recombinant human megakaryocyte growth and development factor as an adjunct to induction and consolidation therapy for patients with acute myeloid leukemia. *Blood.* 2000;95(8): 2530-2535.
74. Dickinson M, Cherif H, Fenaux P, et al. Azacitidine with or without eltrombopag for first-line treatment of intermediate- or high-risk MDS with thrombocytopenia. *Blood.* 2018;132(25):2629-2638.
75. Drexler HG, Quentmeier H. Thrombopoietin: expression of its receptor MPL and proliferative effects on leukemic cells. *Leukemia.* 1996;10(9):1405-1421.
76. Will B, Kawahara M, Luciano JP, et al. Effect of the nonpeptide thrombopoietin receptor agonist eltrombopag on bone marrow cells from patients with acute myeloid leukemia and myelodysplastic syndrome. *Blood.* 2009; 114(18):3899-3908.
77. Sekeres MA, Kantarjian H, Fenaux P, et al. Subcutaneous or intravenous administration of romiplostim in thrombocytopenic patients with lower risk myelodysplastic syndromes. *Cancer.* 2011;117(5):992-1000.
78. Fattizzo B, Levati G, Cassin R, Barcellini W. Eltrombopag in immune thrombocytopenia, aplastic Anemia, and myelodysplastic syndrome: from megakaryopoiesis to immunomodulation. *Drugs.* 2019;79(12): 1305-1319.

© 2026 American Society of Hematology. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.